

Exploring Student Perceptions and Personal Motivations Behind Campus Gym Usage: A Behavioral and Experimental Analysis

Kayden Piniewski, Louise van Dam, Ahnaf Farhan Khan, Katrina Lahey,
Angella Che, and Anastasia Kartadinata

University of Calgary

Data 201: Thinking with Data

Lab: B04, Group 2

Professor: Dr. Andy Asare

TA: Farima Golchin

Submission Date: December 5, 2025

Introduction:

Full-time students at the University of Calgary have busy schedules, sometimes making it difficult to prioritize both physical and mental health. To support students in staying active, our campus gym provides many ways to get moving, such as lifting weights, doing cardio, or working on flexibility and mobility. The gym serves as a central space on campus where students can focus on improving their overall well-being.

However, students' experience with the gym are mixed. Some report that it is too crowded and have had negative experiences while attending. On the other hand, some students have had a great time attending the campus gym because it's easily accessible, convenient, and has a positive atmosphere. We believe this topic about the campus gym is important because the facility fees are included in each student's tuition, with no option for exemption. Therefore, students' opinions about the University of Calgary gym are crucial in creating a positive, inclusive, and useful space for all student body.

The primary objective of our survey is to understand the reasons why students choose to either use or not use our campus gym. We are essentially interested in whether students believe attending the gym enhances their personal lives, including self-image, confidence, and their mental and physical health, as well as how these perceptions influence University of Calgary students' decisions to engage with the facility available to them. Understanding these factors will help us identify methods we can use towards improving the gym experience for all students and aim to encourage more individuals to engage in consistent physical activity to strengthen their overall well-being.

Analysis Questions:

1. What demographic uses the campus gym most frequently (e.g., gender and age)
2. What are the main motivators for students who use the campus gym (e.g., convenience, cleanliness, social reasons)?
3. What are the physical or environmental barriers preventing students from using the campus gym (e.g., crowding, limited hours, or unclean facilities)?
4. What are the psychological or social barriers preventing students from using the campus gym (e.g., fear of judgment, lack of motivation, lack of friends, or low confidence)?
5. How do perceptions of gyms differ between people who use them and those who do not?

Obtaining Data:

Recruitment Message:

Are you a student at the University of Calgary?

We are conducting a short survey about gym usage on campus. The purpose of this survey is to better understand the reasons why students choose to use or not use the campus gym. We aim to better understand students' experiences and perceptions of the gym to identify potential barriers and motivators to participation.

The survey will take approximately 5 minutes to complete, and all responses are completely anonymous. Participation in this study is entirely voluntary, and you may choose to skip any question or withdraw at any time without penalty.

If you are a current student, we invite you to participate and share your feedback.

Exploring Student Perceptions and Personal Motivations behind Campus Gym Usage: A Behavioral and Experimental Analysis

Target Participants:

Our target participants are students who attend the University of Calgary.

The Consent Form:



**UNIVERSITY OF
CALGARY**

Name of Researcher, Faculty, Department, Telephone & Email:

Kayden Piniewski, Faculty of Science, Data Sciences, (415)-523-0069 or kayden.piniewski@ucalgary.ca

Angella Che, Faculty of Arts, Communication and Media Studies, (532)-321-6589 or angella.che@ucalgary.ca

Ahnaf Farhan Khan, Faculty of Science, Computer Sciences, (678)-957-5724 or ahnaf.farhankhan@ucalgary.ca

Katrina Lahey, Faculty of Arts, Communication and Media Studies, (765)-239-9075 or katrina.lahey@ucalgary.ca

Louise Van Dam, Faculty of Arts, Communication and Media Studies, (578)-764-4938 or louise.vandam@ucalgary.ca

Anastasia Kartadinata, Faculty of Arts, Communication and Media Studies, (354)-576-6970 or anastasia.kartadinata@ucalgary.ca

Supervisor:

Dr. Andy Asare

Title of Project:

Exploring student Perceptions and Personal Motivations behind Campus Gym Usage: A Behavioral and Experimental Analysis

Sponsor:

N/A

This consent form, a copy of which has been given to you, is only part of the process of informed consent. If you want more details about something mentioned here, or information not included here, you should feel free to ask. Please take the time to read this carefully and to understand any accompanying information.

The University of Calgary Conjoint Faculties Research Ethics Board has approved this research study.

Participation is completely voluntary, anonymous, and confidential.

Purpose of the Study

The purpose of this study is to explore the reasons why students choose to use or not use the campus gym. We aim to better understand students' experiences and perceptions of the gym to identify potential barriers and motivators to participation.

What Will I Be Asked To Do?

You are asked to complete an online survey for an assignment in an undergraduate university course. The survey is designed to be completed in approximately 10 minutes. The survey is anonymous and does not contain questions asking for specific identifying information.

Examples of survey questions include:

- I use the campus gym because it has a friendly atmosphere*
- I don't use the campus gym because it is too crowded*
- Going to the gym improves overall health and wellbeing*

Participation is completely voluntary. You may refuse to participate altogether, may refuse to participate in parts of the survey, may decline to answer any and all questions, and may withdraw from the survey at any time.

What Type of Personal Information Will Be Collected

No personal identifying information will be collected in this study, and all participants shall remain anonymous.

Should you agree to participate, you will be asked to provide your gender, age, and whether you are currently enrolled at the University of Calgary.

Are there Risks or Benefits if I Participate?

There is no foreseeable risks, harms, or inconveniences to the participant. By participating in the survey, you are helping university undergraduate students to complete a course assignment.

What Happens to the Information I Provide?

Only the students who distribute the survey, as well as the instructor and teaching assistants of the course will have access to the information collected.

Withdrawal of data is not possible after data collection, as all collected data is anonymous and identifiable data cannot be retrieved.

All data collected will be destroyed by the end of the course.

No one except the students who have distributed the survey, the instructor, and the teaching assistants will be allowed to see any of the answers to the survey. There are no names on the survey. Only group information will be summarized for any presentation or publication of results. The survey data will be deleted by the end of the course.

Signatures

Your response below indicates that 1) you understand to your satisfaction the information provided to you about your participation in this survey, 2) you agree to participate in the survey, and 3) you are 18 years of age or older.

In no way does this waive your legal rights nor release the investigators, sponsors, or involved institutions from their legal and professional responsibilities. You are free to withdraw from this survey at any time. You should feel free to ask for clarification or new information throughout your participation.

Questions/Concerns

If you have any further questions or want clarification regarding this survey and/or your participation, please contact:

Kayden Piniewski by email at kayden.piniewski@ucalgary.ca

Angella Che by email at angella.che@ucalgary.ca

Ahnaf Farhan Khan by email at ahnaf.farhankhan@ucalgary.ca

Katrina Lahey by email at katrina.lahey@ucalgary.ca

Louise Van Dam by email at louise.vandam@ucalgary.ca

Anastasia Kartadinata by email at anastasia.kartadinata@ucalgary.ca

If you have any concerns about the way you've been treated as a participant, please contact the Research Ethics Analyst, Research Services Office, University of Calgary at 403.220.6289 or 403.220.8840; email cfreb@ucalgary.ca.

Do you consent to participate?

☐ Yes

☐ No



Screenshots of Survey:

Are you a student or currently a college?

What is your age?

What is your gender?

Do you regularly use the computer at college?

Figure A1

Grouping based on respondent answers

Yes

Rate how much you agree or disagree with each statement:

Using the computer at college is a waste of time.

Using the computer at college is a good use of time.

Using the computer at college is a good use of money.

Using the computer at college is a good use of space.

Using the computer at college is a good use of energy.

Using the computer at college is a good use of resources.

Using the computer at college is a good use of time and space.

Using the computer at college is a good use of time and money.

Using the computer at college is a good use of time and energy.

Using the computer at college is a good use of time and resources.

Figure A2

No

Rate how much you agree or disagree with each statement:

Using the computer at college is a waste of time.

Using the computer at college is a good use of time.

Using the computer at college is a good use of money.

Using the computer at college is a good use of space.

Using the computer at college is a good use of energy.

Using the computer at college is a good use of resources.

Using the computer at college is a good use of time and space.

Using the computer at college is a good use of time and money.

Using the computer at college is a good use of time and energy.

Using the computer at college is a good use of time and resources.

Figure A3

Other

Rate how much you agree or disagree with each statement:

Using the computer at college is a waste of time.

Using the computer at college is a good use of time.

Using the computer at college is a good use of money.

Using the computer at college is a good use of space.

Using the computer at college is a good use of energy.

Using the computer at college is a good use of resources.

Using the computer at college is a good use of time and space.

Using the computer at college is a good use of time and money.

Using the computer at college is a good use of time and energy.

Using the computer at college is a good use of time and resources.

Figure A4

To what extent do you believe each of the following statements is true?

Using the computer at college is a waste of time.

Using the computer at college is a good use of time.

Using the computer at college is a good use of money.

Using the computer at college is a good use of space.

Using the computer at college is a good use of energy.

Using the computer at college is a good use of resources.

Using the computer at college is a good use of time and space.

Using the computer at college is a good use of time and money.

Using the computer at college is a good use of time and energy.

Using the computer at college is a good use of time and resources.

Figure A5

See Appendix A for in-depth details

The data collected will help us determine why students choose to use or avoid the campus gym by examining their experiences and perceptions. By analyzing students' feedback about the gym's atmosphere, we can identify whether environmental factors such as crowding, cleanliness, or inclusivity, influence gym attendance. In addition, understanding students' perceptions and preconceptions about gym use will help reveal potential psychological or social barriers. Together, these insights will allow us to determine the main reasons behind campus gym usage and suggest ways to make the facility more accessible and appealing to all students.

Initial state of dataset (screenshots as examples):

Our dataset was collected through **Qualtrics**, which produced two versions of the data: a *numeric* dataset and a *labeled* dataset. For our project, we chose to work with the *numeric* dataset because it allows smoother statistical analysis, easier grouping, and cleaner visualization for the rest of the team.

Before beginning any analysis, we carefully cleaned and organized the data to ensure accuracy, readability, and consistency in the final data analysis. The following sections explain which attributes were identified as dirty or clean, what cleaning procedures we applied, and why these steps were necessary.

Dirty Attributes

1. Metadata Columns

Why this data needed to be cleaned:

The raw dataset included many metadata columns automatically generated by Qualtrics, none of which helped answer our research questions. Keeping them would clutter the dataset and make analysis more confusing. Removing unnecessary variables improves efficiency and reduces noise during visualization and statistical processing.

Dirty Data Screenshot (before cleaning): Here, red markings are the irrelevant columns we don't need

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	StartDate	EndDate	Status	Progress	Duration (s)	Finished	RecordedDate	RecipientLastN	RecipientF	RecipientE	ExternalRe	Distributio	UserLangu	Q1	Q3	Q2	Q4
2	Start Date	End Date	Respon	Progres	Duratio	Finishe	Recorded Date	Recipient La	Recipie	Recipie	Extern	Distrib	User La	Are you	What is	What is	Do you
3	21-10-2025 17:28	#####	0	100	97	1	#####					anonymo	us	1	20	2	
4	21-10-2025 17:28	#####	0	100	268	1	#####					anonymo	us	1	20	2	
5	21-10-2025 17:47	#####	0	100	93	1	#####					anonymo	us	1	23	1	
6	21-10-2025 17:56	#####	0	100	146	1	#####					anonymo	us	1	20	1	
7	21-10-2025 18:10	#####	0	100	116	1	#####					anonymo	us	1	20	1	
8	21-10-2025 18:11	#####	0	100	127	1	#####					anonymo	us	1	24	2	
9	21-10-2025 18:20	#####	0	100	103	1	#####					anonymo	us	1	21	2	
10	21-10-2025 18:19	#####	0	100	304	1	#####					anonymo	us	1	19	2	
11	21-10-2025 18:28	#####	0	100	136	1	#####					anonymo	us	1	20	2	
12	21-10-2025 19:00	#####	0	100	91	1	#####					anonymo	us	1	20	1	
13	21-10-2025 22:26	#####	0	100	457	1	#####					anonymo	us	1	21	1	
14	21-10-2025 22:38	#####	0	100	122	1	#####					anonymo	us	1	20	2	
15	21-10-2025 22:45	#####	0	100	83	1	#####					anonymo	us	1	22	1	
16	21-10-2025 22:50	#####	0	100	100	1	#####					anonymo	us	1	19	1	
17	21-10-2025 22:56	#####	0	100	87	1	#####					anonymo	us	1	20	2	
18	21-10-2025 22:58	#####	0	100	176	1	#####					anonymo	us	1	20	1	
19	21-10-2025 23:01	#####	0	100	167	1	#####					anonymo	us	1	20	2	
20	21-10-2025 23:51	#####	0	100	109	1	#####					anonymo	us	1	19	2	
21	22-10-2025 00:18	#####	0	100	114	1	#####					anonymo	us	1	20	1	
22	22-10-2025 05:32	#####	0	100	75	1	#####					anonymo	us	1	19	1	
23	22-10-2025 05:36	#####	0	100	100	1	#####					anonymo	us	1	19	1	

Clean Data Screenshot (after cleaning): After deleting columns, the relevant columns needed prevails

SERIAL NO.	(1= YES 2= NO)	AGE	(1= Male 2= Female 3= Non-binary/third gender 4= Prefers not to say)	(1= Yes 2= No 3= Off-Campus)	The Questions And Responses Of Participants Using CAMPUS Gym (1= Strongly Disagree → 5= Strongly Agree)										Responses Made By All Group Of Participants (1= Strongly Disagree → 5= Strongly Agree)										
					REGULARITY OF USING GYM	FIND GYM CONVENIENT	FRIENDS USE GYM	HAS FRIENDLY ATMOSPHERE	HAS GOOD EQUIPMENT	FEEL SAFE & COMFORTABLE	FIND EMPLOYEES APPROACHABLE	USE DUE TO TUITION OBLIGATION	FIND GYM CLEAN	FIND HOURS FIT SCHEDULE	FIND LESS CROWDED	IMPROVES DISCIPLINE	IMPROVES CONFIDENCE	IMPROVES SELF ESTEEM	IMPROVES HEALTH WELLBEING	IS JUST A TREND	UNHEALTHY OBSESSION	DECREASES STRESS	SOCIAL STANDARDS	IMPROVES MENTAL HEALTH	BOOSTS EGO IMAGE
1	1	20	2	1	5	4	3	4	4	4	5	4	5	3	3	3	4	2	4	3	3	4	3		
2	1	21	2	1	5	1	3	2	3	1	4	3	3	1	4	4	3	5	1	3	5	3	3		
3	1	19	2	1	5	4	3	2	4	5	4	4	4	1	4	4	5	5	2	4	4	4	4		
4	1	22	1	1	5	5	5	5	5	5	5	5	5	2	4	5	5	5	3	4	5	1	5		
5	1	20	1	1	5	2	1	2	3	3	5	4	5	1	4	5	5	5	2	2	3	3	4		
6	1	20	2	1	5	1	2	3	4	2	5	4	5	1	4	4	4	5	2	4	4	4	4		
7	1	19	1	1	4	4	4	4	3	1	3	3	1	3	5	5	5	5	1	1	5	4	5		
8	1	27	2	1	5	1	5	5	5	5	2	5	5	5	4	5	5	5	2	2	5	3	5		
9	1	25	2	1	5	2	4	5	4	2	2	4	4	1	3	5	5	5	2	5	4	3	5		
10	2	21	2	1	5	5	1	1	3	3	2	2	5	1	5	5	5	5	2	5	5	3	5		
11	1	20	2	1	5	4	4	4	3	4	4	4	4	1	4	4	3	4	2	4	4	4	3		
12	1	20	2	1	5	1	3	5	4	4	3	3	4	2	4	4	4	4	2	3	3	3	3		
13	1	19	2	1	5	4	3	4	2	5	5	5	5	1	5	5	5	5	2	3	5	5	5		
14	1	19	1	1	5	3	4	5	5	5	3	4	5	3	5	5	5	5	2	2	5	4	5		
15	1	22	1	1	4	5	3	4	2	3	5	4	3	1	5	5	5	5	2	4	5	5	5		
16	1	24	1	1	4	3	3	4	4	4	4	4	3	2	4	4	4	3	3	3	2	3	4		
17	1	21	1	1	5	3	4	4	4	3	5	4	4	2	5	5	5	5	1	4	5	2	4		
18	1	26	2	1	5	3	4	3	4	3	4	3	3	1	5	5	5	5	2	3	5	2	4		
19	1	21	1	1	5	4	4	4	3	3	2	3	4	1	4	4	4	4	1	3	4	3	4		

Cleaning Procedure:

We removed **irrelevant metadata fields**, including Start Date, End Date, Progress, Finished, Status Report, Progress Duration, Recorded Date., Response ID, Recipient Last/First Name, Email, External Reference, Distribution Channel, User Language.

After removing them, the dataset contained only meaningful variables directly related to the gym-usage research topic. So, our dataset is just relevant to our research goals and analysis.

2. Incomplete Responses (< 20% completion)

Why this data needed to be cleaned:

Incomplete surveys produce misleading analysis because missing answers distort averages and visualizations. Participants who answered less than 20% of the survey were not providing reliable or usable data.

101	#####	0	100	74	1	#####	anonym ous	EN	1:20	1	1	5	3	3	2	3	4	3	3	4	1	4	4	4	5	3	2	3	2	4	3							
102	#####	0	100	64	1	#####	anonym ous	EN	1:32	1	3											5	5	5	5	1	2	5	1	5	2							
103	23-10-2025 13:15	0	100	86	1	#####	anonym ous	EN	1:21	1	2											4	4	5	5	1	4	3	2	4	3							
104	23-10-2025 14:12	0	100	226	1	#####	anonym ous	EN	1:24	3	2											2	2	2	3	1	3	2	3	3	2							
105	#####	0	60	9	0	#####	anonym ous	EN	1:18	1	2																											
106	#####	0	100	38	1	#####	anonym ous	EN	1:22	1	3											4	4	4	5	1	5	3	2	4	4							
107	#####	0	100	70	1	#####	anonym ous	EN	1:19	2	3											5	5	5	5	2	4	5	2	5	4							
108	20-10-2025 15:41	0	100	79	1	#####	anonym ous	EN	1:20	1	2											5	4	4	4	2	3	2	5	4	5							
109	#####	0	70	68	0	#####	anonym ous	EN	1:20	2	2																											
110	#####	0	60	9	0	#####	anonym ous	EN	1:18	2	2																											
111	23-10-2025 19:21	0	70	78	0	#####	anonym ous	EN	1:21	1	2																											
112	#####	0	50	92	0	#####	anonym ous	EN	1:23	2	1	5	5	5	5	3	5	3	5	3	3	3	3															
113	#####	0	70	38	0	#####	anonym ous	EN	1:31	2	2																											
114	21-10-2025 10:02	1	100	26	1	#####	proble anonym	EN																														
115	24-10-2025 11:48	0	70	28	0	#####	anonym ous	EN	1:18	1	2																											
116	#####	0	80	15	0	#####	anonym ous	EN	1:20	2	3																											
117	#####	0	80	607	0	#####	anonym ous	EN	1	4	3																											
118	02-11-2025 14:41	0	100	184	1	#####	anonym ous	EN	1:22	1	1	5	5	5	5	5	5	4	5	5	5	4	4	4	5	5	2	1	4	1	4	3						
119	02-11-2025 17:38	0	100	95	1	#####	anonym ous	EN	2:23	1	3											5	5	5	5	1	4	5	1	5	4							
120	#####	0	100	183	1	#####	anonym ous	EN	1:21	2	2											3	3	3	4	1	2	3	2	3	3							
121	03-11-2025 13:23	0	100	100	1	#####	anonym ous	EN	1:20	2	3											5	5	5	5	5	2	4	4	4	4	4	5					
122	#####	0	40	11	0	#####	anonym ous	EN	1:27	2	1																											
123	#####	0	70	76	0	#####	anonym ous	EN	1:21	1	2																											

[illegible]

Identified all surveys with <20% of questions answered and removed them. In total we removed 14 incomplete responses, leaving the data with 107 valid and complete entries.

Why this needed cleaning:

The raw data layout was difficult to read as it had:

- narrow rows
- mismatched column widths
- small text
- plain code like headers (e.g., Q1, Q5_1).

A confusing layout increases the risk of coding errors and slows analysis.

Dirty Data Screenshot (before cleaning):

[illegible]

Clean Data Screenshot (after cleaning):

[illegible]

Cleaning Procedure:

- Adjusted row heights and column widths
- Changed the font to Calibri for readability
- Enlarged and bolded header text
- Added a color scheme to help separate sections and make it easy for eyes
- Added the survey link at the top-left corner for identification
- Inserted two blank rows and columns as per dataset formatting norms
- Changed code names (Q1-5) to the identifying headers based on survey questions

4. Ungrouped Gym Usage Responses

Why this needed cleaning:

The dataset included three gym-usage categories, but responses were scattered. This made comparative analysis (campus gym users vs. non-users vs. off campus users) much harder.

[illegible][illegible]

We sorted participants into three distinct groups based on the question:

- *Yes* $\rightarrow$ Campus Gym Users
- *No* $\rightarrow$ Non-Gym Users
- *Uses Off-Campus Gym* $\rightarrow$ Off-Campus Gym Users

This restructuring made later analysis (visualizations, comparisons, correlations) clear and accurate.

We also merged header cells to label each cluster with both questions and responses of participants:

- Using the campus gym
- Using off-campus gym
- Not using any gym
- Perceptions of the gym (all participants answered these questions)

5. Unclear Numerical Coding (Indicators)

Why this needed cleaning:

Qualtrics exported categories as numbers (e.g., 1 for Male, 2 for Female), but without a visible legend, these numbers were meaningless to the reader. Without clarification, the dataset could easily be misinterpreted.

Dirty Data Screenshot (before cleaning):

01	02	03	04	05-1	05-2	05-3	06-1	06-2	06-3	07-1	07-2	07-3	08-1	08-2	08-3	09-1	09-2	09-3	10-1	10-2	10-3	11-1	11-2	11-3	12-1	12-2	12-3
Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year 16	Year 17	Year 18	Year 19	Year 20	Year 21	Year 22	Year 23	Year 24	Year 25	Year 26	Year 27	Year 28
1.20		2	1	5	4	3	4	4	4	5	4	5	3	3	3	4	2	4	3	3	4	3					

Clean Data Screenshot (after cleaning):

UCALGARY STUDENT STATUS	AGE	(1= Male 2= Female 3= Non-binary/third gender 4= Prefer not to say) GENDER	(1= Yes 2= No 3= Off-Campus) REGULARITY OF USING GYM	The Questions And Responses Of Participants Using CAMPUS Gym (1= Strongly Disagree → 5= Strongly Agree)								
				FIND GYM CONVENIENT	FRIENDS USE GYM	HAS FRIENDLY ATMOSPHERE	HAS GOOD EQUIPMENT	FEEL SAFE & COMFORTABLE	FIND EMPLOYEES APPROACHABLE	USE DUE TO TUITION OBLIGATION	FIND GYM CLEAN	FIND HOURS FIT SCHEDULE

Cleaning Procedure:

Added indicator keys above the dataset:

- Gender: 1 = Male | 2 = Female | 3 = non-binary / third gender | 4 = Prefer not to say
- Student Status: 1 = Yes | 2 = No
- Likert Scale: 1 = Strongly disagree → 5 = Strongly agree
- Likert Scale 1 = None → 5 = A great deal

We merged the Serial No. and Gender columns with their associated headers since they did not require indicator values.

Clean Attributes

These attributes were already clean and did not require additional modifications:

1. Pre-Existing Numeric Values

Why they were clean:

Most survey responses were already converted into numeric codes by Qualtrics. This ensured consistent formatting across the dataset and made statistical analysis straightforward.

What we verified:

- Student status
- Age
- All Likert scale items

No inconsistencies or non-numeric values were detected.

Procedures used to confirm cleanliness:

- Scanned columns for unexpected text or blank entries
- Verified that Likert-scale values were within the correct 1-5 range

2. Consistent Category Coding

Why they were clean:

All “select one” responses used the same numeric system (1, 2, 3...), meaning no recoding errors occurred during export.

Procedures used to verify this:

- Compared variable labels across the dataset
- Confirmed that no categories were duplicated or misaligned

3. Minimal Recoding Required

Why they were clean:

Since Qualtrics had already standardized coding during export, only organizational improvements (headers, grouping, formatting) were needed - not value correction.

Procedures used to verify:

- Checked for contradictions (e.g., out-of-range values)
- Ensured consistent formatting from column to column.

Final Summary

By removing unnecessary fields, filtering incomplete responses, reorganizing the dataset into meaningful groups, simplifying headers, and adding indicator legends, we produced a clean and structured dataset that is easy to interpret and ready for analysis.

Every step of cleaning improved:

Readability, Accuracy, Consistency, Analysis Quality

This cleaned dataset forms a strong foundation for our descriptive statistics, visualizations, and final insights.

Final cleaned state of dataset (screenshots as examples):

[illegible]

Here is a link to all our data excel spreadsheets for more in-depth details:

<https://drive.google.com/drive/folders/1hWODg25TArNjQV3K9Vs1RWhNEoxCLC5Y?usp=sharing>

Visualizations and Findings:

1. What demographic uses the campus gym most frequently (e.g., gender and age)

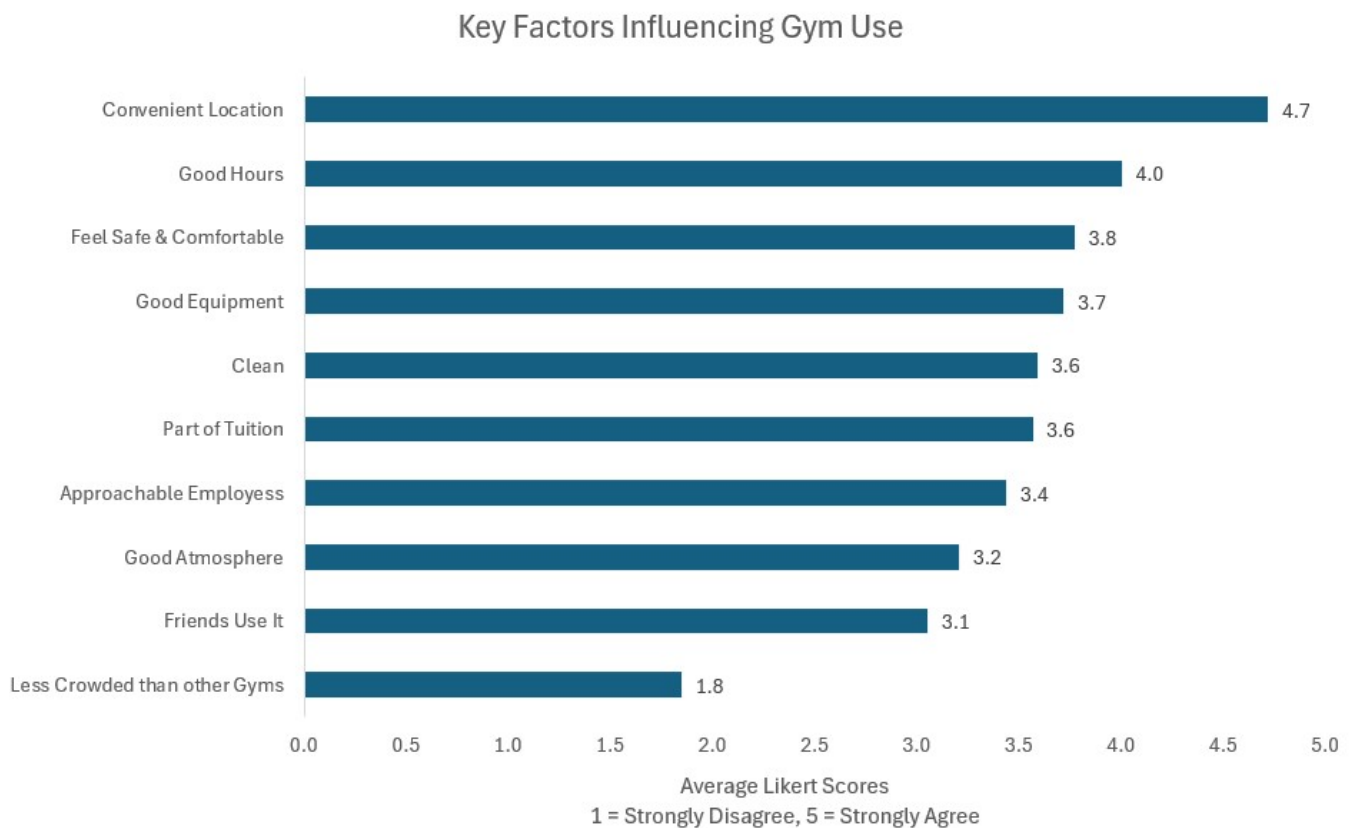
Gym Users by Age and Gender



This visualization uses a sunburst chart because it allows clear comparison across multiple categories at the same time such as gender and age groups. The layered structure makes it easy to see how each age range is distributed within each gender, while also letting the viewer compare the two genders directly.

Overall, men use the campus gym more frequently than women, and the most common age groups are men age 21 and women ages 19–20. This suggests that gym usage is highest among students in their second and third years of university.

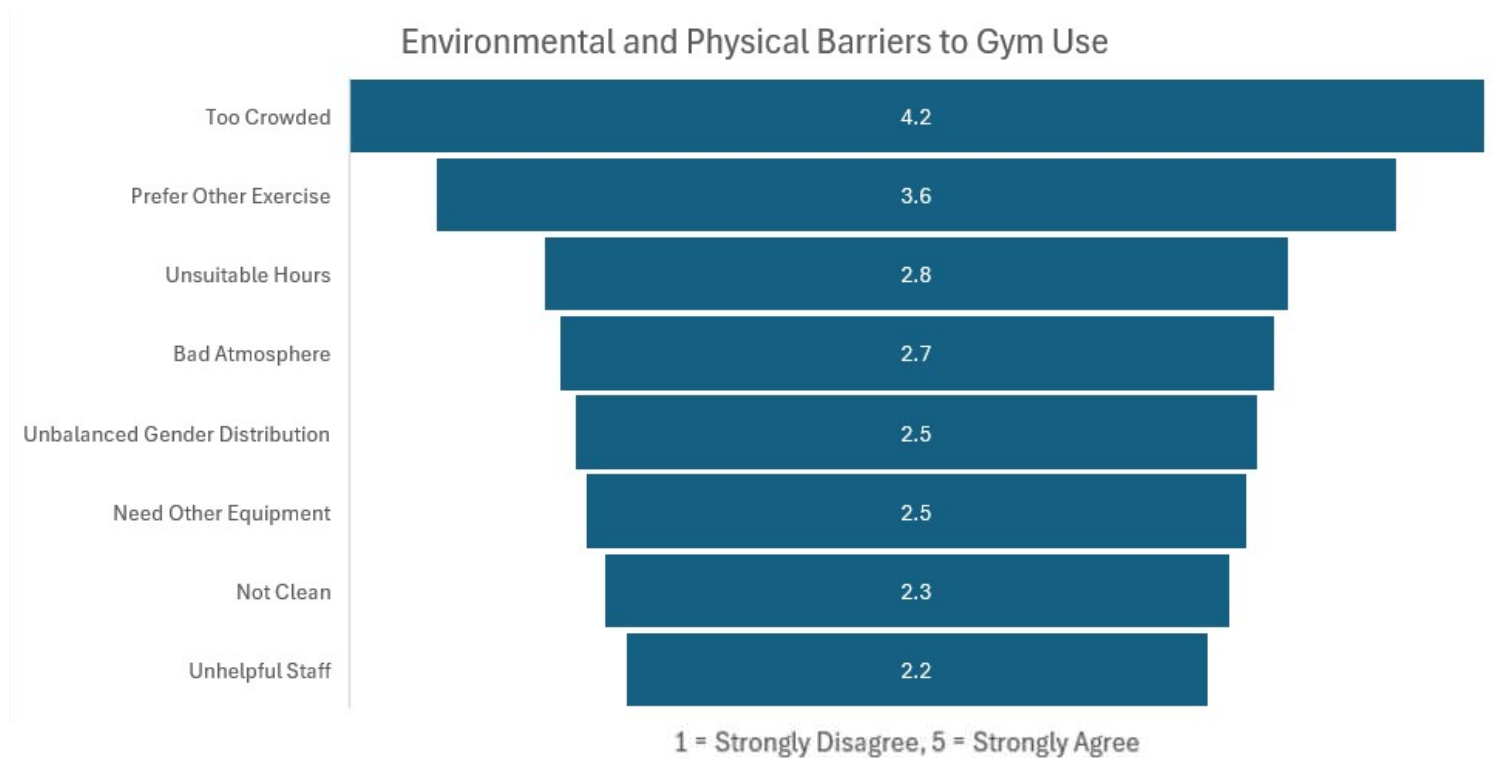
2. What are the main motivators for students who use the campus gym (e.g., convenience, cleanliness, social reasons)?
(scored with a Likert scale measuring agreement)



This visualization uses a clustered bar graph because it makes it easy to compare the different reasons students choose to use the campus gym. The grouped bars clearly show which factors are most influential.

Convenience is the strongest motivator for using the campus gym, followed by its operating hours. Being less crowded than other gyms does not appear to influence attendance, suggesting that students prioritize practical and time-related benefits rather than spatial factors.

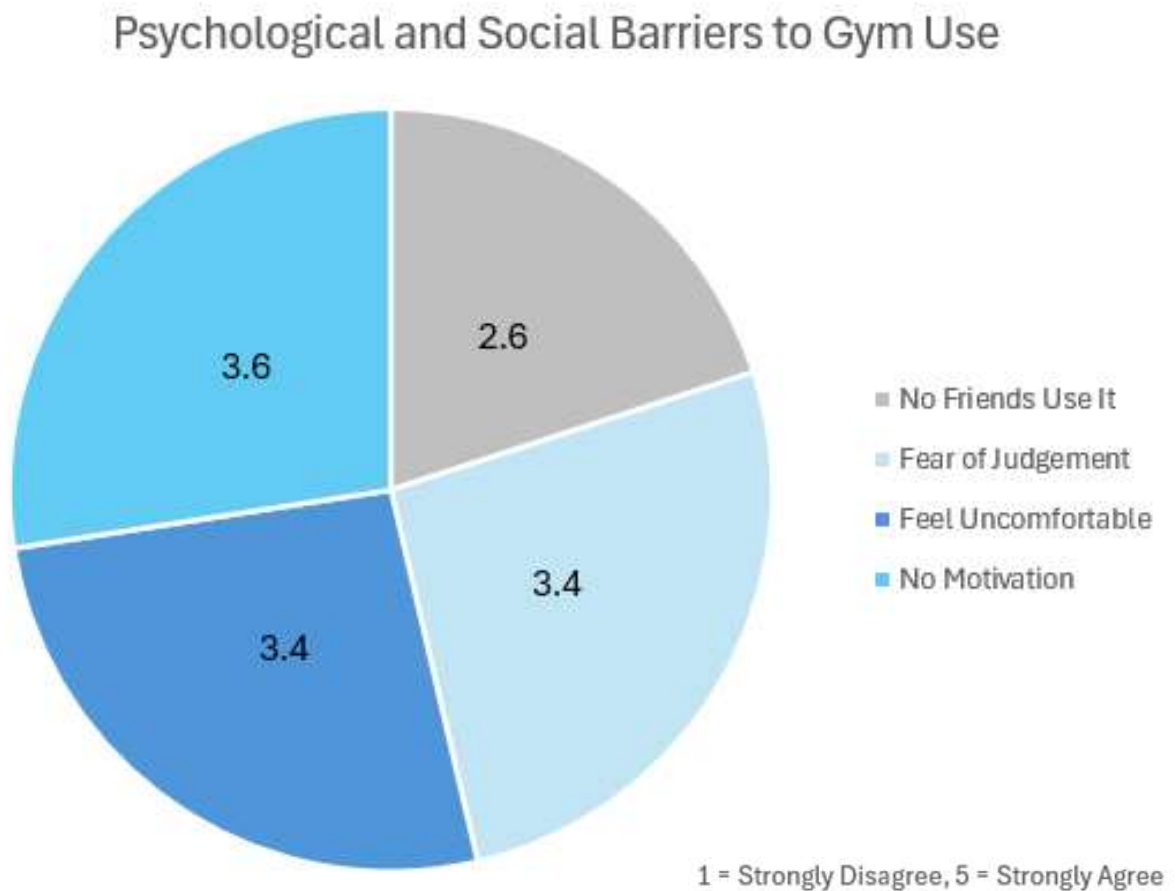
3. What are the physical or environmental barriers preventing students from using the campus gym (e.g., crowding, limited hours, or unclean facilities)?
(scored with a Likert scale measuring agreement)



A funnel graph was used for this data because it clearly shows the relative size of each environmental or physical barrier affecting campus gym usage, highlighting the biggest and smallest factors.

Crowding and preference for other forms of exercise are the most significant environmental barriers. Issues related to staff are the least influential, indicating that physical conditions matter more to users than personnel-related factors.

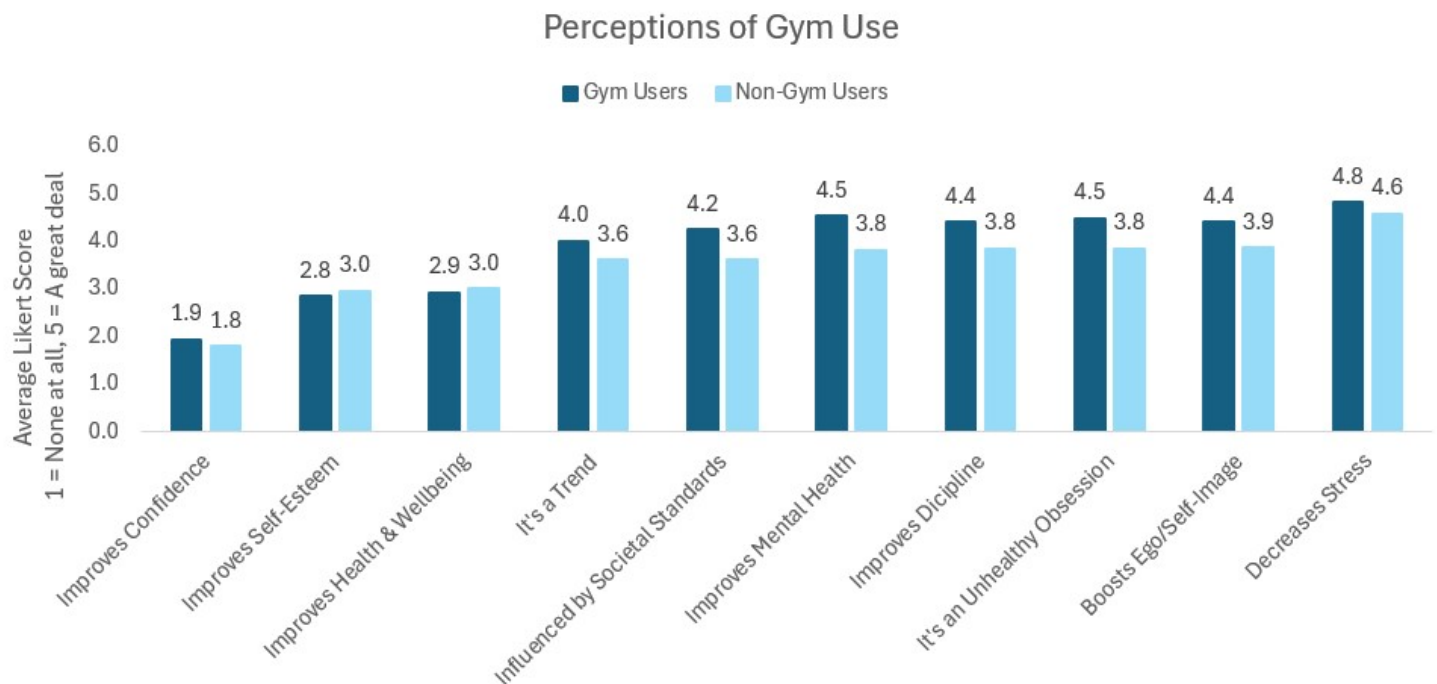
4. What are the psychological or social barriers preventing students from using the campus gym (e.g., fear of judgment, lack of motivation, or low confidence)?
(scored with a Likert scale measuring agreement)



A pie chart was chosen to represent this data because there were only a few factors to compare, and this format helps us easily see which factors influence gym use the most.

Feeling uncomfortable in the gym is the most common psychological barrier, while lack of friends attending the gym is the least. However, all factors score close to neutral, suggesting that psychological barriers overall have a limited effect on gym use.

5. How do perceptions of gyms differ between people who use them and those who do not?
(scored with a Likert scale measuring how much you believe each statement)



This graph is a clustered bar graph as it allows us to compare multiple factors at once, such as the perspectives of people who use the gym versus those who don't.

Both gym users and non-users agree that the gym reduces stress the most, but it has little impact on confidence. Gym users generally rate all benefits higher, suggesting that firsthand experience increases positive perceptions of gym impact.

Contributions:

Introduction and Analysis Questions – Katrina Lahey

Obtaining Data and File Organization – Kayden Piniewski

Data Cleaning– Ahnaf Farhan Khan

Explaining Cleaning Process and Dirty Attributes/Video Editing – Louise van Dam

Visualizations and Findings – Kayden Piniewski (This was assigned to Anastasia, but she didn't do it/did it improperly so Kayden redid all of it).

Interesting Findings and Recommendations – Angella Che

Appendix A

Are you a student at University of Calgary?

☐ Yes

☐ No

What is your age?

What is your gender?

☐ Male

☐ Female

☐ Non-binary / third gender

☐ Prefer not to say

Do you regularly use the gym on campus?

☐ yes

☐ no

☐ I attend a gym off campus



Figure A1

Rate how much you agree or disagree with each statement.

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I use the campus gym because it's convenient	☐	☐	☐	☐	☐
I use the campus gym because my friends use it	☐	☐	☐	☐	☐
I use the campus gym because it has a friendly atmosphere	☐	☐	☐	☐	☐
I use the campus gym because it has a good selection of equipment	☐	☐	☐	☐	☐
I use the campus gym because I feel safe and comfortable there	☐	☐	☐	☐	☐
I use the campus gym because the employees are approachable	☐	☐	☐	☐	☐
I use the campus gym because I am forced to pay for it in my tuition	☐	☐	☐	☐	☐
I use the campus gym because it is clean	☐	☐	☐	☐	☐
I use the campus gym because the hours work better with my schedule	☐	☐	☐	☐	☐
I use the campus gym because it's less crowded than other gyms	☐	☐	☐	☐	☐



Figure A2

Rate how much you agree or disagree with each statement.

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I don't use the campus gym because it is too crowded or over-stimulating	☐	☐	☐	☐	☐
I don't use the campus gym because it is not clean or well-maintained	☐	☐	☐	☐	☐
I don't use the campus gym because the equipment does not fit my needs	☐	☐	☐	☐	☐
I don't use the campus gym because I feel uncomfortable or intimidated there	☐	☐	☐	☐	☐
I don't use the campus gym because I prefer to exercise in other ways (sports, outdoors, at home)	☐	☐	☐	☐	☐
I don't use the campus gym because I worry others will judge me	☐	☐	☐	☐	☐
I don't use the campus gym because I don't have the motivation	☐	☐	☐	☐	☐
I don't use the campus gym because the hours do not fit my schedule	☐	☐	☐	☐	☐
I don't use the campus gym because the staff are not approachable or helpful	☐	☐	☐	☐	☐
I don't use the campus gym because I don't have friends to go with	☐	☐	☐	☐	☐



Figure A3

Rate how much you agree or disagree with each statement.

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I don't use the campus gym because it is too crowded	☐	☐	☐	☐	☐
I don't use the campus gym because it is not clean or well-maintained	☐	☐	☐	☐	☐
I don't use the campus gym because the equipment does not fit my needs	☐	☐	☐	☐	☐
I don't use the campus gym because I feel uncomfortable or intimidated there	☐	☐	☐	☐	☐
I don't use the campus gym because my friends use a different one	☐	☐	☐	☐	☐
I don't use the campus gym because it does not fit my schedule	☐	☐	☐	☐	☐
I don't use the campus gym because the staff are not approachable	☐	☐	☐	☐	☐
I don't use the campus gym because the atmosphere is not welcoming	☐	☐	☐	☐	☐
I don't use the campus gym because the gender distribution is not balanced	☐	☐	☐	☐	☐
I don't use the campus gym because it has limited hours	☐	☐	☐	☐	☐



Figure A4

To what extent do you believe each of the following statements is true

	None at all	A little	A moderate amount	A lot	A great deal
Going to the gym improves discipline.	☐	☐	☐	☐	☐
Going to the gym improves confidence.	☐	☐	☐	☐	☐
Going to the gym improves self-esteem.	☐	☐	☐	☐	☐
Going to the gym improves overall health and wellbeing.	☐	☐	☐	☐	☐
Going to the gym is just a trend.	☐	☐	☐	☐	☐
Going to the gym can become an unhealthy obsession.	☐	☐	☐	☐	☐
Going to the gym decreases stress levels	☐	☐	☐	☐	☐
Going to the gym is influenced by societal standards	☐	☐	☐	☐	☐
Going to the gym improves mental health	☐	☐	☐	☐	☐
Going to the gym boosts ego/self-image.	☐	☐	☐	☐	☐



Figure A5